

SAYALI MOGHE

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PROFESSIONAL EXPERIENCE

Software Engineer

June 2023 – Present

Empyra

Dallas, TX

- **Pitched and led development of a natural-language database interface (MCP server + GitHub Copilot)**, securing leadership approval and shipping it to production for **4 clients** – enabling non-technical users to query data conversationally while enforcing guardrails against exposing schema, SQL, or PII.
- **Owned client-facing delivery of 25 compliance reports** for government workforce-agency clients – independently gathering requirements directly with stakeholders, reviewing and approving report logic built by an offshore team before client release, maintaining **100% on-time delivery**.
- Designed and built a **flexible, config-driven workflow engine** that triggers API calls to external systems on user actions, data-state changes, or scheduled events – without requiring code changes per integration; **powers 15+ workflows across 2 B2B integrations**, cutting new-integration build time from weeks to days.
- Architected a **distributed, multi-tenant Elasticsearch** search platform (BM25 ranking, async indexing, batch processing) supporting **~75** production instances, improving search relevance by **~71%** and indexing performance by **~67%**.
- **Built a personalized job-matching and recommendation system** (12 REST API endpoints, Elasticsearch-backed search) for a client-facing job-search feature, integrating real-time filtering with profile-based recommendation logic.
- **Executed ETL data-migration pipelines** (Python, SQL Server) for **3+ enterprise clients**, processing a **56GB legacy dataset** and migrating **5M+ records** into production with **zero downtime**.
- **Built a custom AI agent** that auto-generated technical documentation for **over 400 regulatory data fields** by parsing the codebase and reference material, cutting documentation time from an estimated 2 months to 2 weeks (**~75% faster**).
- Resolved **50,000+ data inconsistencies** in a regulated benefits-attribution system by designing a validation framework and reprocessing **700K+ historical records** to retroactively apply corrected business logic.

Machine Learning Engineer

October 2022 – May 2023

Miko

Pittsburgh, PA

- Led a 6-person team to **build a speech synthesis pipeline** using **Whisper large-v2**, generating high-quality timestamped audio clips (**≤10s**) and curated **22.05kHz** datasets via MOS-based filtering.
- **Fine-tuned VITS model via ESPNet** toolkit for text-to-speech, enabling **10+ dynamic personalized voices** and developed a cross-singer voice conversion model (SVC) for robotics applications.

PROJECTS

Hybrid Book Recommendation System

- Engineered hybrid recommendation engine combining collaborative filtering (ALS) and semantic embeddings (384-dim sentence-transformers), achieving validation NDCG@5 of 0.39 across 14.7k items and 26.8k users.
- Designed FastAPI backend serving REST endpoints, CORS middleware, and error handling; built recommendation API and asynchronous ML pipeline orchestration with real-time progress tracking for long-running jobs.

Question Answering Platform

- Built a BERT-based QA system using HuggingFace Transformers, SpaCy for NER & coreference resolution, and Docker for containerized deployment; integrated with REST API for scalable query handling.

EDUCATION

Master of Science in Artificial Intelligence and Innovation

Carnegie Mellon University (CMU) - School of Computer Science

August 2021 – May 2023

Pittsburgh, PA

Bachelor of Engineering in Computer Science

University of Mumbai (MU)

August 2017 – June 2021

Mumbai, India

SKILLS

Languages & Databases: Python, C#, Java, SQL, MS SQL Server, PostgreSQL

Frameworks & Libraries: .NET Core, FastAPI, React, Vue, Vite

Cloud, DevOps & Data Infra: AWS, Docker, Kubernetes, Jenkins, Apache Spark, Kafka, Elasticsearch, Logstash, Git, CI/CD

AI/ML & Data Engineering: GitHub Copilot, HuggingFace Transformers, OpenAI API, scikit-learn, SpaCy, LangChain, Pandas, NumPy